

Technical Memorandum, Part 1: Capital Bikeshare Demand Analysis & Forecasting

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TO: Dr. Sohn's Research Lab, PhD Technical Assessment Review
FROM: Sampada Kharel (sampadha.kharel@gmail.com)
DATE: September 22, 2026
RE: Part 1 findings: descriptive statistics, regression analysis, and demand forecasting (full code: *part1_bikeshare.ipynb*)

Data & Methodology. This analysis uses the UCI Bike Sharing dataset (*hour.csv*, 17,379 hourly records, Washington D.C., 2011–2012). Weather variables were denormalized using the UCI web page's min-max formulas (temp: -8 – 39°C ; atemp: -16 – 50°C). **Season codes were corrected** to match the observed calendar months (1=winter...4=fall) rather than the assessment brief's labels, which are off by one step. The dataset is missing 165 of 17,544 possible hours; because no observed row has count = 0 and gaps cluster at night and during Hurricane Sandy, the January 2011 snowstorm, and Hurricane Irene, missing hours are treated as unrecorded zero-rental hours for feature construction; the main analysis and all reported metrics use observed hours only.

Task 1: Descriptive Statistics & Exploratory Analysis

- **Overdispersion:** mean hourly count ≈ 189 , variance $\approx 32,900$. Variance far exceeds the mean for count, casual, and registered rentals, ruling out a simple Gaussian/Poisson-equidispersed model.
- **Hour $\times$ workingday interaction:** workdays show a sharp bimodal commute pattern (peaks at 08:00 and 17–18:00); weekends/holidays show one broad midday peak: two qualitatively different daily shapes.
- **Casual vs. registered:** casual (leisure) demand is far more weekend- and weather-sensitive (weekend mean 58.7 vs. weekday mean 26.3 rentals/hr); registered (commuter) demand peaks on workday commute hours. Casual share of total rentals swings from $\approx 9\%$ at commute hours to $\approx 26\%$ in the afternoon.
- **Year-over-year growth:** mean demand rose from 143.8 (2011) to 234.7 (2012) rentals/hour, a 63% increase, so any forecasting model must represent this trend.
- **Data-quality anomalies:** weather category 4 has only 3 hours; windspeed = 0 in 12.5% of hours (likely a sensor floor, not true calm air); a single day (2011-03-10) has humidity = 0 (sensor fault).

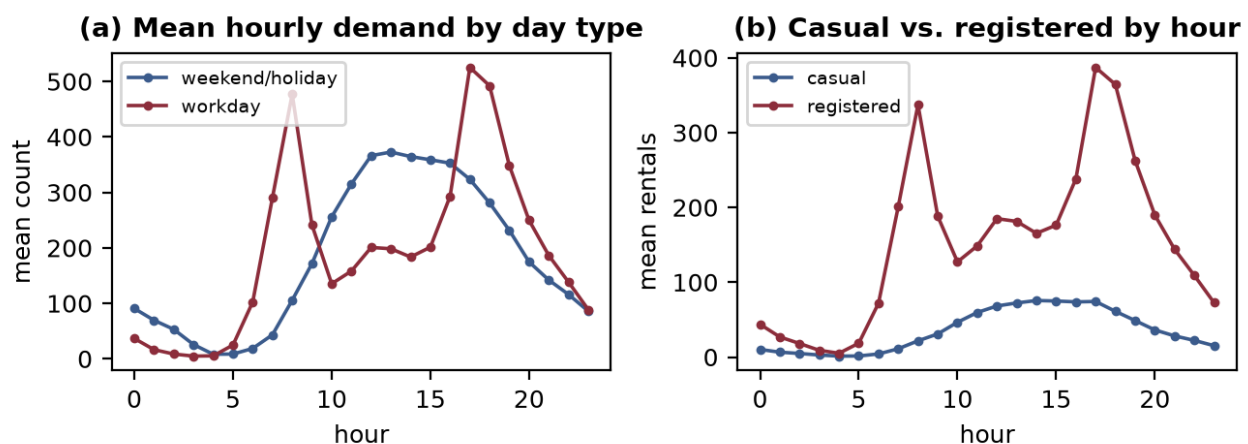


Figure 1. (a) Mean hourly demand by day type; (b) casual vs. registered rentals by hour.

Task 2: Statistical & Regression Analysis

Models were built in stages to isolate each statistical issue named in the assessment: an OLS model on $\log(\text{count}+1)$ exposes multicollinearity and non-linearity; moving to Poisson, then Negative Binomial, GLMs addresses the count-data

distribution directly.

Multicollinearity, non-linearity, and overdispersion

Diagnostic	Result	Action taken
VIF: temp vs. atemp	43.6 / 43.7 (both) → 1.0 (temp alone)	Dropped atemp
Non-linearity (temp)	AIC 31,260 → 30,846 adding temp ²	Quadratic term retained (p<0.001)
Hour × workingday	AIC 30,846 → 14,755 adding interaction	Interaction retained (F-test p≈0)
Overdispersion test	Cameron-Trivedi est. $\alpha=0.041$, $t=35.0$, $p=0$	Negative Binomial (2-step) preferred over Poisson
NB model fit	AIC: Poisson 327,627 vs. NB2 175,749	NB2 used for count
Residual autocorrelation	Durbin-Watson = 0.83; HAC/naive SE ratio ≈1.26×	HAC (Newey-West, 24-lag) SEs reported

Note: the joint-MLE Negative Binomial failed to converge on this design (too many sparse dummy cells); the Cameron-Trivedi two-step estimator (Poisson → auxiliary OLS for α → GLM-NB with fixed α) was used instead and converges cleanly.

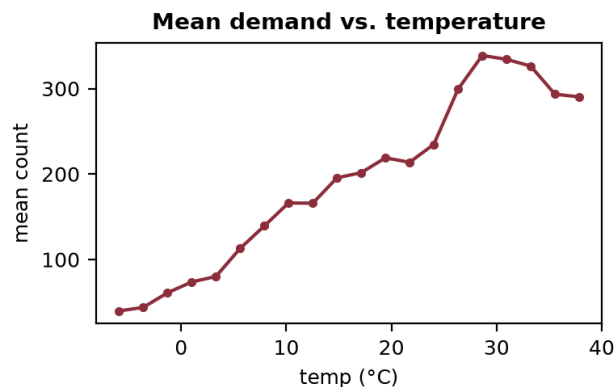


Figure 2. Demand rises with temperature, then flattens/declines, motivating the quadratic term.

Casual vs. registered: incidence rate ratios (selected terms, Negative Binomial)

Term	Casual IRR	p	Registered IRR	p
workingday	0.45	<0.001	0.99	0.092
holiday	0.82	<0.001	0.79	<0.001
temp (°C)	1.18	<0.001	1.06	<0.001
weather = adverse (3)	0.53	<0.001	0.62	<0.001
year = 2012	1.36	<0.001	1.63	<0.001

- Workday status roughly **halves casual demand** (IRR 0.45) but has a small, non-significant marginal effect on registered demand once hour is controlled for. The registered commute signature lives in *which hours* are busy (the hour×workingday interaction above), not the daily level.
- Casual demand is more weather-sensitive (IRR 0.53 vs. 0.62 in adverse weather) and grew more slowly year-over-year (IRR 1.36 vs. 1.63) than registered demand, consistent with membership-driven growth.
- Variable selection combined domain reasoning, nested-model AIC/F-tests (above), and a LassoCV robustness check that retained the same features (temp, temp², hour, and weather dummies dominate by magnitude).

Task 3: Demand Forecasting & Predictive Modeling

Split: train = days 1–20 of each month (11,460 hours); test = day 21–month end (5,919 hours; 24 monthly windows), as prescribed. **Leakage design:** the longest test window is 264 hours, so any feature lag under 264 hours can reference a

timestamp inside the very window being predicted, verified both analytically and with an executable check. All history-derived features use a 336-hour (2-week) lag or longer, and a demonstration confirms a naive 24-hour lag would leak into 5,448 of 6,024 test hours, while 336 hours leaks into none. **Two scenarios** are reported: (A) strict, calendar + history features only (realistic case); (B) oracle weather, adds true future weather as an upper bound. **Features:** hour/day/month/year, cyclical encodings, workingday/holiday/rush-hour flags, a day-index trend, and three leakage-safe history features (a 336h lag, a 4–7 week average, and an expanding same-hour-of-week mean).

Model comparison (test-set RMSLE, primary metric)

Model	Scenario	RMSLE	MAE	RMSE
Seasonal-mean baseline	A (strict)	0.651	79.4	118.9
Poisson GLM	A	0.580	54.0	90.2
Random Forest	A	0.536	47.3	80.5
LightGBM (Poisson)	A	0.531	47.5	81.5
Poisson GLM	B (oracle weather)	0.472	41.1	68.8
Random Forest	B	0.472	38.8	66.3
LightGBM (Poisson)	B	0.440	34.3	59.2

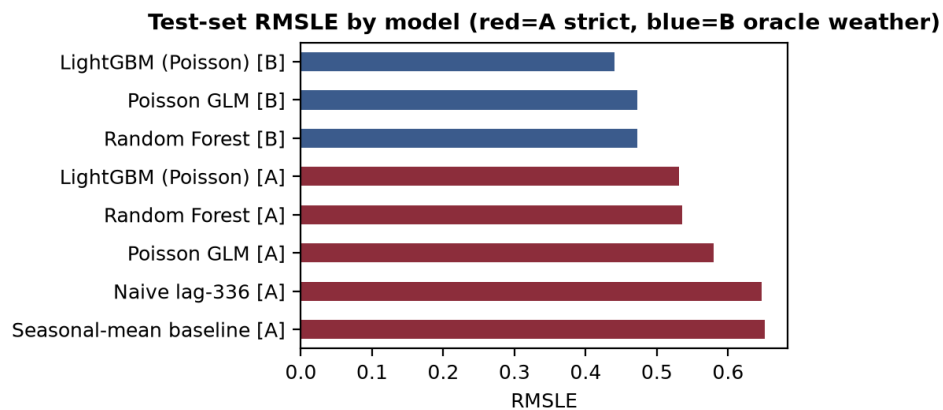


Figure 3. Test RMSLE by model and scenario.

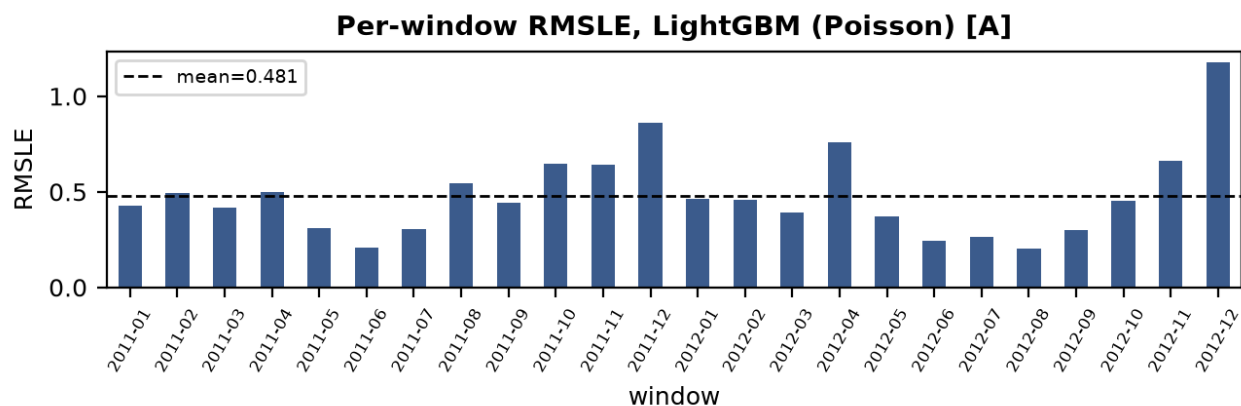


Figure 4. Per-window RMSLE, best strict model (LightGBM, Scenario A).

- LightGBM (Poisson objective) is the best model in both scenarios, beating the seasonal-mean baseline by $\approx 18\%$ and the naive 2-week-lag baseline similarly on RMSLE.
- Knowing future weather (Scenario B) improves RMSLE by a further $\approx 17\%$ ($0.531 \rightarrow 0.440$), quantifying how much realistic-case error is attributable to weather uncertainty rather than the model itself.

- **Worst-performing windows** (Dec 2012, RMSLE 1.18; Dec 2011; Nov 2012) coincide with the largest within-month demand-level shifts (e.g., Dec 2012 mean demand fell 63% from the training to the test portion of the month), not with the Sandy/Irene storm windows. A strong correlation ($r = 0.75$) links per-window error to this level shift. History-based features cannot anticipate a seasonal regime change.
- A fully sequential **rolling-origin backtest** (training only on data prior to each window) gives RMSLE 0.544, close to the prescribed split's 0.531, supporting the headline result despite the prescribed split's known chronology quirk (it pools all months' first 20 days regardless of order).

Limitations & Conclusion

- Missing-hour imputation (zero rentals) is inferred, not documented in the source; a sensitivity check excluding these hours from feature history was not exhaustively re-verified beyond the checks in Section 3.
- The prescribed train/test split allows later months to inform earlier test windows; the rolling-origin backtest addresses this but was only evaluated for the single best Scenario-A model.
- No classical time-series model (SARIMAX/Prophet) was fit as a further comparison, given time constraints; the tree/GLM comparison presented is considered sufficient to support the conclusions above.

Overall, the dataset shows strong, well-understood seasonal and behavioral structure (commute vs. leisure usage, overdispersed counts, temperature non-linearity), which a Negative Binomial GLM explains well for inference, and which a leakage-safe LightGBM model forecasts with a realistic RMSLE of 0.53, a meaningful, verified improvement over naive baselines, with weather uncertainty and seasonal regime shifts identified as the two main remaining error sources.